Lesson 1: Money Sense: Rounding, Estimating and Using Your Calculator

Assignment

1. Use the amount \$31.857.
(a) Which digit is in the tenths place? _____
(b) Which digit is in the hundredths place? _____
(c) Which digit is in the thousandths place? _____
(d) What is the value of the 8? _____

2. Write each amount in dollars.
(a) nine dollars and six cents _____
(b) 40¢ _____
(c) 5¢ _____
(d) gas at 152.9¢ per litre _____ per litre
(e) Put these prices in order from least to greatest: \$3.09 \$3.9 \$3.19 \$3.91

3. Round each amount.

Amount	Nearest dollar	Nearest ten cents	Nearest cent
\$7.384			
\$12.659			
\$0.847			
\$49.995			

4. A calculator shows each answer below. Round it to the nearest cent and write it as money.
(a) `23.4567` _____
(b) `17.2849` _____
(c) `104.995` _____
(d) `0.666666667` _____

5. Malik buys school supplies for \$3.79, \$1.29, \$6.49 and \$2.85.
(a) Find a front-end estimate of the total.

(b) Find the exact total.

(c) Malik has \$15. Is that enough?

6. Ali is making spaghetti for his family.

Item	Price
Spaghetti	\$2.29
Pasta sauce	\$3.79
Ground beef	\$8.64
Lettuce	\$2.49
Parmesan cheese	\$6.19
Garlic bread	\$3.50

(a) Round each price to the nearest dollar and estimate the total.

(b) Find the exact total.

(c) Ali pays with \$30 in cash. How much change does he get?

7. For each calculation, estimate the answer. Then decide if the calculator answer is reasonable. If it is not, find the correct answer.

Calculation	**Calculator shows**	**My estimate**	**Reasonable? If not, correct it.**
(a) $6 \times \$2.15$	`129`		
(b) $\$48.60 \div 4$	`12.15`		
(c) $\$19.99 + \$24.50 + \$9.75$	`54.24`		
(d) $4 \times \$38.95$	`15.58`		

8. Work out each by hand, using BEDMAS.
(a) $5 + 3 \times 4$ (b) $(5 + 3) \times 4$ (c) $20 - 12 \div 4$ (d) $(20 - 12) \div 4$

9. At the movies, Hana buys 2 popcorns at \$8.75 each and 3 drinks at \$5.25 each.
(a) Write one calculation for the total cost. _____
(b) Find the total with your calculator.

(c) Check your answer with an estimate.

10. Four friends order a pizza for \$32.97 and wings for \$18.47. They share the cost equally.
(a) Write one calculation with brackets, and find each person's share.

(b) Jaden types 32.97 [+] 18.47 [÷] 4 [ENTER] and gets `37.5875`. What went wrong?

11. Priya works at a grocery store for \$16.40 per hour.
(a) Last week she worked 22.5 hours. Estimate her pay, then find her exact pay.

(b) How many hours does she need to work to earn \$500? Round to the nearest tenth of an hour.

12. A 12-pack of pop costs \$8.49.
(a) What does one can cost, to the nearest cent?

(b) What does one can cost, to the nearest tenth of a cent?

13. Making change.

(a) A smoothie costs \$7.35. You pay with a \$10 bill. Count up to find your change, and list the coins you get.

(b) A pair of jeans costs \$26.80. You pay with a \$50 bill. What is your change?

(c) Write what you pay for each total if you pay with cash.

Total	\$18.43	\$7.38	\$31.06	\$4.91
Cash price				

(d) For the \$18.43 total, what is your change from a \$20 bill if you pay cash?

14. Three friends go out for supper. They order a burger for \$17.95, a pasta for \$19.50, a salad for \$14.25 and 3 drinks at \$3.25 each. Together they have \$60.

(a) Estimate the bill. Is the estimate close to \$60?

(b) Write one calculation for the exact bill and find it.

(c) Do they have enough money?

(d) A parent gives them money to make up the difference. Now they split the bill equally. How much does each friend pay?

15. \$24.546 rounded to the nearest ten cents is

A. \$24.50 **B.** \$24.55 **C.** \$24.60 **D.** \$25.00

16. Three people share two bills, \$45.60 and \$29.85, equally. Which keys give each person's share?

A. 45.60 [+] 29.85 [÷] 3 [ENTER]

B. [(] 45.60 [+] 29.85 [)] [÷] 3 [ENTER]

C. 45.60 [÷] 3 [+] 29.85 [ENTER]

D. [(] 45.60 [+] 29.85 [)] [×] 3 [ENTER]

17. Numerical Response A 6-pack of yogurt costs \$5.49. The cost of one yogurt, to the nearest cent, is \$_____.

18. Numerical Response Ava buys 2 notebooks at \$3.45 each and 4 pens at \$1.15 each. She pays with a \$20 bill. Her change, in dollars, is _____.

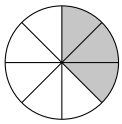
Lesson 2: Fractions Refresher: Equivalent Fractions and Simplest Form

Assignment

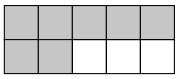
1. Write the fraction of each picture that is shaded.

(a) 

Shaded: _____

(b) 

Shaded: _____

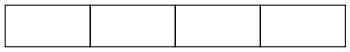
(c) 

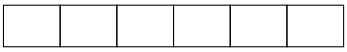
Shaded: _____

(d) 

Filled: _____

Shade each bar to show the fraction.

(e) $\frac{3}{4}$ 

(f) $\frac{2}{3}$ 

2. Write three fractions equivalent to each fraction.

(a) $\frac{1}{2}$

(b) $\frac{3}{5}$

(c) $\frac{12}{18}$ (one smaller and one bigger)

3. Find the missing number.

(a) $\frac{2}{5} = \frac{\square}{15}$

(b) $\frac{3}{8} = \frac{\square}{40}$

(c) $\frac{21}{28} = \frac{3}{\square}$

(d) $\frac{4}{9} = \frac{28}{\square}$

(e) $\frac{36}{48} = \frac{\square}{4}$

(f) $\frac{10}{15} = \frac{\square}{12}$ (simplify first)

4. Complete the table.

Fraction	**Factors of the top**	**Factors of the bottom**	**GCF**	**Simplest form**
$\frac{12}{18}$				
$\frac{30}{70}$				
$\frac{16}{40}$				
$\frac{27}{45}$				

5. Write each fraction in simplest form.

(a) $\frac{6}{8}$ (b) $\frac{9}{12}$ (c) $\frac{14}{21}$ (d) $\frac{20}{36}$

(e) $\frac{35}{50}$ (f) $\frac{48}{64}$ (g) $\frac{45}{60}$ (h) $\frac{75}{100}$

6. Sofia was asked to write $\frac{24}{40}$ in simplest form. She wrote $\frac{24}{40} = \frac{12}{20}$.

(a) Is $\frac{12}{20}$ equal to $\frac{24}{40}$? _____

(b) Is $\frac{12}{20}$ in simplest form? Explain, and give the correct answer.

7. A delivery driver made 45 deliveries on Friday. 36 of them were on time.

(a) What fraction of the deliveries were on time? Write it in simplest form.

(b) What fraction of the deliveries were late? Write it in simplest form.

8. Complete the table. Round repeating decimals to the nearest thousandth.

Fraction	$\frac{3}{4}$	$\frac{2}{5}$	$\frac{5}{8}$	$\frac{1}{3}$			
Decimal					0.35	0.8	0.875

9. Write $<$, $>$ or $=$ in each box. Show your method.

(a) $\frac{3}{4} \square \frac{5}{8}$ (b) $\frac{2}{3} \square \frac{7}{10}$ (c) $\frac{9}{12} \square \frac{6}{8}$ (d) $\frac{4}{9} \square \frac{5}{11}$

10. A carpenter has drill bits in these sizes (in inches): $\frac{3}{16}$, $\frac{1}{4}$, $\frac{5}{32}$, $\frac{3}{8}$. Put them in order from smallest to largest.

11. In Ms. Okafor's class, 18 of 24 students passed the first-aid test. In Mr. Lam's class, 21 of 30 students passed. Which class had the greater fraction of students pass? Explain.

12. Find each amount. Round money to the nearest cent.

(a) $\frac{2}{3}$ of 27 (b) $\frac{3}{5}$ of 40 (c) $\frac{5}{6}$ of 90 (d) $\frac{7}{10}$ of \$85 (e) $\frac{3}{8}$ of \$59

13. Amira earns \$480 every two weeks at her part-time job. She puts $\frac{1}{4}$ of her pay into savings and spends $\frac{3}{8}$ of her pay on her car.

(a) How much does she put into savings?

(b) How much does she spend on her car?

(c) How much of her pay is left?

(d) What fraction of her pay is left? Write it in simplest form.

14. A 20 L water jug for a soccer team is $\frac{3}{4}$ full before practice. During practice, the team drinks $\frac{2}{5}$ of the water that was in the jug. How many litres of water are left in the jug?

15. Which fraction is *not* equivalent to $\frac{3}{4}$?

A. $\frac{6}{8}$ **B.** $\frac{15}{20}$ **C.** $\frac{5}{6}$ **D.** $\frac{30}{40}$

16. Which fraction is the largest?

A. $\frac{5}{8}$ **B.** $\frac{3}{5}$ **C.** $\frac{7}{12}$ **D.** $\frac{2}{3}$

17. Numerical Response Jordan's phone plan has 16 GB of data each month. He has used $\frac{5}{8}$ of it. How many gigabytes has he used?

18. Numerical Response Write $\frac{9}{16}$ as a decimal, rounded to the nearest hundredth.

Lesson 3: Ratios

Assignment

1. A parking lot has 8 trucks, 5 cars and 3 vans. Complete the table.

Comparison	**Colon**	**Words**	**Fraction**	**Part-to-part or part-to-whole?**
trucks to cars				
cars to trucks				
vans to all vehicles				
cars to all vehicles				

2. A hockey team has 13 forwards, 7 defence players and 2 goalies. Write each ratio in simplest form.

(a) defence to forwards

(b) goalies to the whole team

(c) forwards to players who are not forwards

3. Write each ratio in simplest form.

(a) 6 : 9 (b) 20 : 5 (c) 16 : 24 (d) 18 : 18 (e) 35 : 14 (f) 48 : 60

4. Write each ratio in simplest form, using whole numbers.

(a) 1.5 : 3.5 (b) 0.4 : 1.2 (c) 2.25 : 0.75 (d) 4.8 : 3.6

(e) A bank's board shows the example rate 1 CAD = 0.75 USD. Write the ratio of Canadian dollars to US dollars in simplest form, using whole numbers.

5. Change to the same unit, then write each ratio in simplest form.

(a) 30 min : 1 h (b) 600 mL : 2 L (c) 75 cm : 3 m (d) 750 g : 2 kg (e) 20 s : 2 min

(f) Jordan says the ratio of 40 min to 1 h is 40 : 1. Explain his mistake and give the correct ratio in simplest form.

6. Priya's phone has 128 GB of storage. She has used 48 GB.

(a) Write the ratio of used storage to free storage in simplest form.

(b) Write the ratio of used storage to total storage in simplest form.

7. A fruit punch recipe uses 3 L of juice for every 2 L of ginger ale. Complete the ratio table.

Juice (L)	3	6		15	
Ginger ale (L)	2		8		1

8. Which ratios are equivalent to 4 : 6? Circle all that are.

(a) 2 : 3 (b) 8 : 12 (c) 6 : 4 (d) 10 : 15 (e) 12 : 16

9. A granola recipe uses oats, nuts and raisins in the ratio 4 : 2 : 1 (by cups).

(a) Sam uses 6 cups of oats. How many cups of nuts and raisins does he need?

(b) How many cups of granola does Sam make?

(c) Write the ratio of raisins to the whole mix.

10. Tara mixes 5 L of white paint with 2 L of blue paint. Dev mixes 8 L of white paint with 3 L of blue paint. Do they get the same shade? If not, whose paint is lighter (more white for each litre of blue)? Explain.

11. Share each amount in the given ratio.

(a) \$40 in the ratio 3 : 5

(b) \$210 in the ratio 2 : 5

12. A greenhouse worker mixes plant food and water in the ratio 1 : 7. She fills a 12 L watering tank. How much plant food and how much water does she use?

13. Riya and her cousin Matteo split the cost of a game console in the ratio 2 : 3. Riya pays the smaller share, \$180. What did the console cost? How much did Matteo pay?

14. A job site has 12 apprentices and 15 journeypersons. The ratio of apprentices to *all* workers on the site, in simplest form, is

A. 4 : 5 **B.** 4 : 9 **C.** 5 : 9 **D.** 9 : 4

15. The ratio 1 h : 20 min in simplest form is

A. 1 : 20 **B.** 3 : 1 **C.** 1 : 3 **D.** 60 : 20

16. Amir and Lena share \$84 in the ratio 3 : 4. The larger share is

A. \$28 **B.** \$36 **C.** \$48 **D.** \$63

17. Numerical Response The ratio 45 min : 2.5 h in simplest form is $3 : b$. The value of b is _____.

18. Numerical Response A bag of trail mix has peanuts, raisins and chocolate chips in the ratio 5 : 3 : 2 by mass. The bag holds 800 g. The mass of raisins, in grams, is _____.

Lesson 4: Proportions: Finding the Missing Value

Assignment

1. Do the two ratios form a proportion? Show how you know.

(a) $3 : 5$ and $9 : 15$ (b) $\frac{4}{6}$ and $\frac{10}{12}$ (c) $7 : 2$ and $21 : 6$ (d) $\frac{8}{10}$ and $\frac{12}{15}$

2. Solve each proportion by scaling. Write the $\times$ or $\div$ you used.

(a) $\frac{2}{7} = \frac{x}{28}$ (b) $\frac{5}{6} = \frac{30}{x}$ (c) $\frac{24}{32} = \frac{x}{4}$

(d) $\frac{x}{10} = \frac{27}{30}$ (e) $4 : 9 = 16 : x$ (f) $45 : x = 9 : 2$

3. Solve by cross-multiplying. Round to the nearest tenth where needed.

(a) $\frac{x}{8} = \frac{15}{12}$ (b) $\frac{9}{x} = \frac{6}{10}$ (c) $\frac{7}{12} = \frac{x}{30}$

(d) $\frac{13}{x} = \frac{5}{9}$ (e) $\frac{x}{4.5} = \frac{6}{7.5}$ (f) $\frac{11}{15} = \frac{40}{x}$

4. Tyler solved $\frac{4}{x} = \frac{10}{25}$ like this: $4 \times 10 = 25 \times x$, $40 = 25x$, $x = 1.6$.

(a) What mistake did Tyler make?

(b) Solve the proportion correctly.

5. Check each student's answer using cross products. If the answer is wrong, find the correct one.

(a) Mai says the solution of $\frac{6}{14} = \frac{x}{35}$ is $x = 15$.

(b) Devon says the solution of $\frac{x}{9} = \frac{10}{12}$ is $x = 8$.

6. A chili recipe serves 8 people. It uses 900 g of ground beef and 2 cans of beans.

(a) How much ground beef is needed to serve 6 people?

(b) How many cans of beans are needed to serve 20 people?

7. On a road map of Alberta, 1 cm represents 25 km.

(a) Calgary and Red Deer are 5.8 cm apart on the map. About how far apart are they really?

(b) Calgary and Edmonton are about 280 km apart. How far apart are they on the map?

8. On a floor plan for a basement, 1.5 cm represents 2 m.

(a) A family room is 6.3 cm long on the plan. How long is the real room?

(b) A real wall is 3.5 m long. How long should it be on the plan? Round to the nearest tenth of a centimetre.

9. Ella's chainsaw needs gas and oil mixed in the ratio 40 : 1.

(a) How many millilitres of oil should she add to 2 L of gas?

(b) She has a 100 mL bottle of oil. How many litres of gas can she mix with all of it?

10. A medicine is ordered at 25 mg for every 2 kg of body mass. *(Practice numbers only.)*

(a) How many milligrams should a 34 kg child get?

(b) Another patient was given 600 mg. What is the patient's mass?

11. A photo from a phone is printed 12 cm wide and 9 cm tall.

(a) It is enlarged to 40 cm wide without stretching. How tall is the enlargement?

(b) A small copy for a wallet is 6 cm tall. How wide is it?

12. One box of laminate flooring covers 2.2 m^2. Nadia is putting laminate in a 38 m^2 basement room.

(a) How many boxes must she buy?

(b) Each box costs $54.99. What do the boxes cost before tax?

13. A hot chocolate mix says to use 3 tbsp of powder for every 250 mL of milk. Kofi has 2 L of milk and 20 tbsp of powder.

(a) Does he have enough powder to use all the milk? Explain.

(b) How much milk can he use with his 20 tbsp of powder? Round to the nearest tenth of a litre.

14. A recipe uses 4 cups of flour for every 3 cups of sugar. Which proportion finds the flour needed for 5 cups of sugar?

A. $\frac{4}{3} = \frac{x}{5}$ **B.** $\frac{4}{3} = \frac{5}{x}$ **C.** $\frac{3}{4} = \frac{x}{5}$ **D.** $\frac{4}{5} = \frac{x}{3}$

15. What is the solution of $\frac{8}{x} = \frac{20}{35}$?

A. 87.5 **B.** 23 **C.** 14 **D.** 4.6

16. Numerical Response A pool needs 30 mL of chlorine for every 4000 L of water. A pool holds 52 000 L of water. The number of millilitres of chlorine it needs is _____.

17. Numerical Response The solution of $\frac{x}{7} = \frac{18}{11}$, to the nearest tenth, is _____.

Lesson 5: Rates and Unit Rates

Assignment

1. Decide whether each statement is a rate or a ratio.

(a) 3 cups of juice to 5 cups of water _____ (b) 240 km in 3 h _____

(c) $9.60 for 4 L of milk _____ (d) 12 cm to 20 cm _____

(e) 45 students for 2 buses _____ (f) 6 wins to 4 losses _____

2. Write each as a rate, as a fraction and with a slash. Include the units.

(a) Rosa earns $96 for 6 h of babysitting.

(b) A tap drips 15 L of water every 2 h.

(c) A savings account earned $18.40 interest in 4 months.

(d) 1 cm on a trail map stands for 2 km.

3. Find each unit rate. Include the units.

(a) 210 km in 3 h

(b) $78 for 4 h of work

(c) 345 words in 5 min

(d) 2.4 kg of dog food in 8 days

(e) 1.6 km in 8 min

(f) 90 L of water in 2.5 min

4. Sofia earns $136.50 for 7 h of work at a coffee shop.

(a) What is her hourly wage?

(b) How much will she earn in a month when she works 32 h?

(c) How many hours must she work to earn $468.00?

5. Marcus types 55 words/min.

(a) How many words can he type in 12 min?

(b) How long will it take him to type a 1430-word report?

6. Shaved ham costs \$1.98 per 100 g at a deli in Airdrie.
(a) How much do 300 g cost?

(b) Hana spends \$9.90 on ham. How many grams does she get?

7. A landscaper mixes 45 mL of liquid fertilizer into 3 L of water.
(a) How many millilitres of fertilizer is that per litre of water?

(b) How much fertilizer does she need for a 12 L sprayer tank?

8. A machine in a bakery makes 168 cookies in 12 min. How many cookies can it make in a 6-hour shift?

9. One 1 L tin of deck stain covers 9 m^2. Owen's deck has an area of 52 m^2.
(a) How many tins should he buy? Explain.

(b) Each tin costs \$24.99. What will the tins cost (before tax)?

10. Find the fuel consumption of each vehicle in L/100 km.
(a) A car used 37.8 L of gas to drive 540 km.

(b) A work truck used 68.4 L of gas to drive 570 km.

(c) Which vehicle uses less fuel? Explain.

11. Grace is driving from Calgary to Edmonton and back. It is about 300 km each way. Her car uses 8.5 L/100 km and gas costs \$1.39/L.
(a) How many litres of gas will she need for the round trip?

(b) How much will the gas cost?

(c) Her gas tank holds 50 L. Can she make the round trip on one full tank? Explain.

12. Kayla has two job offers. The grocery store pays \$151.20 for an 8-hour shift. The movie theatre pays \$96.25 for a 5-hour shift.
(a) Find the hourly wage for each job.

(b) Complete: The _____ pays more because _____.

(c) How much more would Kayla earn at the better-paying job in a 20-hour week?

13. Convert each rate.
(a) 3.6 km/h to m/min (b) 150 m/min to km/h (c) \$21.60/h to dollars per minute

14. Ella says her car uses 0.065 L of gas per kilometre.
(a) Write her car's fuel consumption in L/100 km.

(b) Explain why fuel consumption is given per 100 km instead of per 1 km.

15. Ana earns \$171.00 for 9 h of work. At the same rate, how much will she earn for 14 h?
A. \$0.74 **B.** \$185.00 **C.** \$266.00 **D.** \$2394.00

16. A speed of 9 km/h is the same as
A. 0.15 m/min **B.** 150 m/min **C.** 540 m/min **D.** 9000 m/min

17. Numerical Response A nurse counts a patient's pulse and gets 21 beats in 15 s. The patient's heart rate, in beats per minute, is _____.

18. Numerical Response A car uses 6.8 L/100 km. The amount of gas it needs for a 350 km trip, to the nearest tenth of a litre, is _____ L.

Lesson 6: Unit Price: Finding the Best Buy

Assignment

1. Find each unit price. Round to the nearest cent.

(a) 4 toothbrushes for $9.96

(b) a 10 kg bag of rice for $21.90

(c) 8 cans of pop for $5.99

(d) 5 L of windshield washer fluid for $4.49

2. A box of 12 granola bars costs $5.88.

(a) Estimate the price of one bar.

(b) Calculate the price of one bar.

(c) Mateo's calculator shows 2.04. What mistake did he make? How does your estimate help you see it is wrong?

3. A sports store buys a box of 50 hockey pucks for $62.50. It sells the pucks for $2.99 each.

(a) How much does one puck cost the store?

(b) How much profit does the store make on one puck?

(c) How much profit does the store make if it sells the whole box?

4. Which is the better buy? Show the unit price of each.

(a) 6 rolls of paper towel for $8.49, or 12 rolls for $15.99

(b) a 340 g bag of coffee for $8.99, or a 925 g can for $24.99

5. Olive oil comes in three sizes.

Size	Price	Work	Per 100 mL
500 mL	\$7.99		
1 L	\$13.49		
3 L	\$37.99		

(a) Complete the table.

(b) Which size is the best buy?

_____

(c) Give one reason, **not** price, why someone might still choose the 1 L bottle.

6. Ana is putting trim around a room. The store sells the trim in 2.4 m pieces for \$7.68 each, or in 3.6 m pieces for \$11.16 each.

(a) Find the price per metre of each piece. Which is the better buy?

(b) Ana needs exactly 7.2 m of trim. Find the cost using only short pieces, and using only long pieces. How much does she save with the better choice?

7. Change to the same unit, then decide which is the better buy.

(a) ketchup: 750 mL for \$4.29, or 1.25 L for \$6.49

(b) apples: \$4.39 per kg loose, or a 1.36 kg bag for \$5.49

(c) grapes: a 500 g bag for \$3.49, or \$6.59 per kg loose

8. A can of soup costs \$1.49. This week the sign says "3 for \$4.00".

(a) What is the price of one can with the deal?

(b) About how much do you save on each can with the deal?

(c) Sam buys 9 cans at the deal price. How much does he save compared with paying \$1.49 for each?

9. At the arena canteen, a sports drink costs \$2.75. At a grocery store, a case of 24 of the same drinks costs \$27.36.

(a) Find the price of one drink in the case.

(b) A team needs 24 drinks. How much does the team save by buying the case instead of buying 24 drinks at the

canteen?

10. Lena runs a small bakery. She orders cake boxes from one of two suppliers.

- Supplier A: 100 boxes for \$48.00, plus \$12.50 delivery.
- Supplier B: 150 boxes for \$69.00, plus \$18.00 delivery.

Which supplier is the better buy, and by how much per box?

11. A loaf of bread costs \$3.49. A two-pack of the same bread costs \$5.99. Devon lives alone and eats about one loaf every two weeks. The bread goes mouldy in about one week.

(a) Find the price per loaf in the two-pack.

(b) Which should Devon buy? Finish the sentence.
Devon should buy _____ because _____

_____

(c) What could Devon do so that the two-pack becomes a good choice?

12. (a) A 2.5 kg bag of carrots costs \$3.45. At the same unit price, how much would 4 kg cost?

(b) Bulk almonds cost \$1.86 per 100 g. Complete the price chart.

50 g	250 g	350 g	1 kg

13. A daycare uses 3 L of milk each day. It is open 5 days a week. Milk costs \$6.20 for a 4 L jug.

(a) How many litres of milk does the daycare use in one week?

(b) What is the price of milk per litre?

(c) How much should the daycare budget for milk for 4 weeks?

14. (a) Rubber gym flooring costs \$12.40 per m^2. Jordan's garage gym is 3 m by 4.5 m. How much will the flooring cost?

(b) A sheet of plywood is 1.2 m by 2.4 m and costs \$46.08. Find the price per m^2.

15. A 400 g jar of salsa costs \$3.60. The store also sells a 650 g jar.

(a) What is the highest price the 650 g jar can have and still not cost more per 100 g than the 400 g jar?

(b) The 650 g jar costs \$5.79. Is it the better buy? Show how you know.

16. A pack of 6 muffins costs \$5.34. The price of one muffin is

A. \$0.89 **B.** \$1.12 **C.** \$32.04 **D.** \$0.53

17. Which size of juice is the best buy?

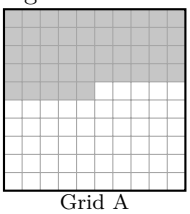
A. 250 mL for \$1.99 **B.** 500 mL for \$3.79 **C.** 1 L for \$7.89 **D.** 2 L for \$15.49

18. Numerical Response Large-flake oats cost \$0.63 per 100 g. The cost of 1.5 kg of oats, to the nearest cent, is \$_____.

Lesson 7: Percents Refresher: Percents, Decimals and Percent of an Amount

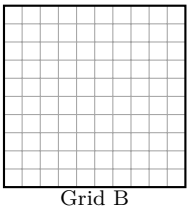
Assignment

1. Each grid is one whole (100 squares).



Grid A

(a) For Grid A, write the shaded part as a percent _____, a decimal _____ and a fraction in simplest form _____. What percent is *not* shaded? _____



Grid B

(b) Shade 8% of Grid B. Write 8% as a decimal. _____

2. Write each percent as a decimal.

(a) 42% _____ (b) 9% _____ (c) 60% _____ (d) 3.5% _____

(e) 150% _____ (f) 0.8% _____ (g) 210% _____ (h) 100% _____

3. Write each decimal or fraction as a percent.

(a) 0.37 _____ (b) 0.9 _____ (c) 0.04 _____ (d) 1.3 _____

(e) 0.125 _____ (f) $\frac{3}{5}$ _____ (g) $\frac{7}{20}$ _____ (h) $\frac{5}{8}$ _____

4. Complete the table. Write fractions in simplest form. Round the percent to the nearest tenth and the decimal to the nearest thousandth where needed.

Percent	**Decimal**	**Fraction**
30%		
	0.85	
		$\frac{1}{4}$
2%		
		$\frac{2}{3}$
	1.75	

5. (a) Complete the table without a calculator.

Amount	**10%**	**5%**	**15%**	**20%**	**25%**	**50%**	**1%**
\$70.00							
\$12.60							

(b) Estimate a 15% tip on a bill of $59.40 in your head. Show your thinking.

6. Calculate. Estimate first to check your answer.

(a) 45% of 80 (b) 12% of 450 (c) 6.5% of 200 (d) 175% of 48

(e) 0.6% of 3500 (f) 300% of 19 (g) 15% of $27.80 (h) 2.5% of $1240

7. A first-year apprentice electrician earns 60% of a journeyperson's wage. The journeyperson earns $48.50 per hour.

(a) How much does the apprentice earn per hour?

(b) How much does the apprentice earn for a 40-hour week?

8. Four friends have pizza at a restaurant in Calgary. The bill is $82.00 and they leave an 18% tip.

(a) Find the tip.

(b) Find the total with the tip.

(c) They split the total equally. How much does each friend pay?

9. Find each percent. Round to the nearest tenth of a percent if needed.

(a) 18 of 24 (b) 36 of 60 (c) 7 of 40

(d) 90 of 75 (e) 5 of 12 (f) 13 of 7

10. Jordan got 34 out of 40 on Quiz A and 41 out of 50 on Quiz B.

(a) Write each score as a percent.

(b) Jordan says he did better on Quiz B because he got more marks. Is he right? Explain.

11. Sam says, "4% as a decimal is 0.4, so 4% of \$250 is \$100." Explain Sam's mistake and find the correct answer. Use a benchmark to show that \$100 cannot be right.

12. Ava's paycheque is \$640.00. She puts 15% into savings and 35% toward her car payment.

(a) How much goes into savings?

(b) How much goes toward the car?

(c) How much is left? What percent of her paycheque is that?

13. 10% of a number is 7.4.

(a) What is the number?

(b) Find 35% of the number using benchmarks (no calculator). Check with a calculator.

14. As a decimal, 0.5% is

A. 0.5 **B.** 0.05 **C.** 0.005 **D.** 5

15. A 20% tip on a bill of \$43.50 is

A. \$8.70 **B.** \$4.35 **C.** \$52.20 **D.** \$0.87

16. 18 is what percent of 15?

A. 83.3% **B.** 120% **C.** 18% **D.** 1.2%

17. Numerical Response 7.5% of 84 is _____.

18. Numerical Response Ty answered 43 out of 52 questions correctly on a driver's test practice quiz. His score, as a percent to the nearest tenth, is _____%.

Lesson 8: Setting a Price: Markup and Sales Tax

Assignment

1. Complete the table.

Rate	25%	5%	13%	150%	7.5%
As a decimal					
Multiply by $(1 + \text{rate})$					

2. Find the markup and the retail price.

(a) cost \$20.00, markup 50%

(b) cost \$64.00, markup 25%

(c) cost \$8.40, markup 120%

(d) cost \$135.60, markup 35%

3. A corner store buys a case of 24 energy drinks for \$27.60. It marks them up 85%.

(a) What does the store pay for one can?

(b) What is the retail price of one can?

4. Find the markup percent. Round to the nearest tenth of a percent if needed.

(a) cost \$40.00, retail \$50.00

(b) cost \$12.50, retail \$30.00

(c) cost \$3.20, retail \$7.99

5. (a) A streaming service raises its monthly price from \$16.99 to \$18.99. What is the percent increase, to the nearest tenth?

(b) A hair salon raises the price of a \$42.00 haircut by 6%. What is the new price?

6. A plumber buys a hot water tank for $725.00. She charges her customer 18% more than she paid.
(a) What is her markup in dollars?

(b) What does she charge the customer for the tank (before tax)?

7. These prices are in Alberta. Estimate the GST first, then find the GST and the total.

Price	Estimate of GST	GST	Total
(a) $24.99			
(b) $8.49			
(c) $1299.00			

8. Snowboard boots cost $219.99 in Kelowna, BC (5% GST and 7% PST). Find the GST, the PST and the total cost.

9. A hotel room in Halifax, Nova Scotia costs $164.00 a night. Nova Scotia has 14% HST. What is the total for 2 nights?

10. An e-bike costs $1849.00 at a store in Calgary. The same e-bike costs $1749.00 at a store in Vancouver, BC (12% tax in all).
(a) Which store is cheaper after tax, and by how much?

(b) Give one reason, other than price, that a person might still buy the e-bike in Vancouver.

11. A sports store buys hockey sticks for $72.00 each and marks them up 65%.
(a) Find the retail price.

(b) What does a customer in Calgary pay, with GST?

(c) If the store were in Toronto (13% HST), what would the customer pay?

12. Complete this receipt from Airdrie, AB. The customer pays with a $110.00 gift card. How much is left on the card?

Nose Creek Hardware Airdrie, AB			
Item	**Qty**	**Price each**	**Amount**
Paint brush	3	$6.49	
Roller cover	2	$8.99	
Paint, 3.78 L	1	$54.98	
Painter's tape	1	$7.49	
		Subtotal	
		GST 5%	
		Total	
		Gift card	$110.00
		Left on card	

13. Jada says, "A 20% markup and then 5% GST is the same as adding 25% to the cost." Test her idea with an item that costs $100.00. Is she right? Explain.

14. Two stores in Edmonton sell the same jacket. Store A pays $60.00 for it and marks it up 75%. Store B pays only $56.00 but marks it up 90%. Which store charges the customer less, with GST, and by how much?

15. In Alberta, the total cost of a $40.00 item with GST is

A. $2.00 **B.** $40.05 **C.** $42.00 **D.** $60.00

16. A store pays $25.00 for a shirt and sells it for $35.00. The markup percent is

A. 10% **B.** 28.6% **C.** 40% **D.** 140%

17. Numerical Response The GST on a $68.40 purchase in Alberta, to the nearest cent, is $_____.

18. Numerical Response A store buys a backpack for $14.40 and sells it for $24.99. The markup percent, to the nearest tenth of a percent, is _____%.

Lesson 9: On Sale! Discounts and Sale Prices

Assignment

1. Complete the table.

Percent off	20%	35%	50%	8%	12.5%	75%
Percent you pay						
Multiply by						

2. Estimate first, then calculate the discount and the sale price.
(a) A backpack, regularly \$49.99, 20% off (b) A car seat, regularly \$219.00, 15% off

3. Use the shortcut (multiply by the percent you pay) to find each sale price.
(a) A bike helmet, \$64.00, 25% off (b) A skateboard deck, \$89.95, 40% off (c) A laptop, \$1149.99, 10% off

4. A print shop charges \$0.18 per page. Orders of 500 pages or more get 15% off. What does an order of 800 pages cost before tax?

5. At a "Hockey Weekend" sale, skates are 25% off, sticks are 30% off and tape is 10% off. Owen buys skates (\$229.96), a stick (\$119.90) and 3 rolls of tape at \$4.50 each. What is his total before tax?

6. Find the percent decrease.
(a) \$50.00 to \$35.00 (b) \$240.00 to \$180.00 (c) A used car: \$12 400 to \$10 850

(d) A drill goes on sale from \$90.00 to \$72.00. Jesse says it is 25% off because $18 \div 72 = 0.25$. Explain his mistake

and give the correct percent.

7. In Red Deer, a microwave is regularly \$159.00. It is on sale for 20% off. Find the sale price, the GST (5%) and the total cost.

8. On a trip to Toronto, Mei buys sneakers that are regularly \$140.00 at 35% off. Ontario charges 13% HST.

(a) What is her total cost?

(b) How much less is this than the regular price with HST?

9. A sports store sells cans of tennis balls for \$7.99 each with the deal Buy 3, get 1 free.

(a) What do you pay for 4 cans? What percent do you save?

(b) Another store has the same cans at \$7.99 with "Buy one, get the second half price". What percent do you save there?

(c) Both deals save the same percent. Give a reason you might choose one over the other.

10. The same speaker is sold at two stores. Store A: regularly \$129.99, 25% off. Store B: regularly \$119.99, \$20 off.

(a) Which store has the lower price, and by how much?

(b) Give one reason, other than price, that someone might still buy at Store B.

11. A clearance rack says "40% off, plus an extra 25% off at the till". A sweater is regularly \$80.00.

(a) What is the final price?

(b) Natalie thinks this is 65% off. What is the real total percent off?

12. A garden store buys a patio umbrella at a wholesale price of $45.00 and marks it up 60%. At the end of summer it sells the umbrella at 50% off.

(a) What was the regular (retail) price?

(b) What is the end-of-summer sale price?

(c) What does a customer in Alberta pay on sale, with 5% GST?

(d) Did the store make or lose money on this umbrella (before tax)? How much?

13. A sofa cost $899.00 in August. In October the store raised the price to $1099.00, then ran an ad: "25% off. Save big!"

(a) What is the sale price?

(b) Compared with the August price, what percent do you really save? (Nearest tenth of a percent.)

14. A $45.00 item is on sale for 30% off. The sale price is

A. $13.50 **B.** $31.50 **C.** $15.00 **D.** $58.50

15. A store takes 20% off, then an extra 20% off the sale price. This is the same as a single discount of

A. 40% **B.** 36% **C.** 64% **D.** 20%

16. Numerical Response A bike is regularly $650.00 and is on sale for $494.00. The percent decrease, to the nearest whole percent, is _____%.

17. Numerical Response A store takes 25% off, then an extra 10% off the sale price. The total percent off, to the nearest tenth, is _____%.

Lesson 10: Currency Exchange: Converting, and Buy vs Sell Rates

Assignment

1. Use 1 EUR = 1.55 CAD.

(a) Which is worth more, one euro or one Canadian dollar? _____

(b) Complete the table.

EUR	1	2	10	40	100
CAD					

2. The rate is 1 CAD = 0.725 USD. Change each amount into US dollars.

(a) $20.00 CAD (b) $260.00 CAD (c) $1450.00 CAD

3. The rate is 1 USD = 1.38 CAD. Find each US price in Canadian dollars.

(a) A movie ticket, $15.00 USD (b) A pizza slice, $8.79 USD (c) A used snowmobile, $2300.00 USD

4. Kai changes $100.00 CAD into two different currencies.

(a) The rate is 1 CAD = 0.549 GBP. Will he get more or fewer than 100 pounds? How many?

(b) The rate is 1 CAD = 13.3 MXN. Will he get more or fewer than 100 pesos? How many?

(c) Complete: If 1 CAD equals a number **less than 1** of a currency, then that currency is worth _____ than a Canadian dollar.

5. Only one rate is posted in each part. Decide whether to multiply or divide.

(a) The rate is 1 GBP = 1.82 CAD. How many pounds do you get for $300.00 CAD?

(b) The same rate. A scarf in Edinburgh costs 18.00 GBP. What is that in CAD?

(c) The rate is 1 CAD = 13.3 MXN. A tour in Cancún costs 900 MXN. What is that in CAD?

6. Use this rate table to convert each amount. Show your multiplication.

Currency	**1 unit = _____ CAD**	**1 CAD = _____ units**
USD	1.36	0.735
EUR	1.52	0.658
MXN	0.078	12.8
JPY	0.0091	110

(a) $180.00 CAD into pesos

(b) 3500 JPY into CAD

(c) $420.00 CAD into yen

(d) 86.00 EUR into CAD

(e) $39.95 USD into CAD

7. Marcus is a millwright apprentice. His shop orders a torque wrench from a US supplier for $184.50 USD. The rate is 1 USD = 1.38 CAD.

(a) What does the wrench cost in Canadian dollars?

(b) His boss set a budget of $250.00 CAD for the wrench. Is it under budget? By how much is it over or under?

8. The rate is 1 USD = 1.38 CAD. Estimate each price in CAD in your head (add about a third), then find the exact price.

Item	**Estimate (CAD)**	**Exact working**	**Exact (CAD)**
(a) T-shirt, $36.00 USD			
(b) Jeans, $90.00 USD			
(c) Concert ticket, $150.00 USD			
(d) Phone charger, $12.99 USD			

(e) Give one reason why it is useful to estimate prices in your head when you shop in the US.

(f) Jordan says, “$70.00 USD is about $51 CAD.” Use an estimate to show he is wrong. What did he do, and what is the right answer?

9. Use this bank's board (CAD for 1 unit).

Currency	We buy	We sell
USD	1.33	1.43
EUR	1.49	1.61
MXN	0.069	0.081

(a) Amira is buying euros for a trip to Spain. Which rate does the bank use: the buying rate or the selling rate? Why?

(b) How much does she pay in CAD for 350 EUR?

(c) Her brother buys 6000 MXN for a trip to Puerto Vallarta. How much does he pay in CAD?

10. Use the board from Question 9. After a trip to Mexico, Devon has 2200 MXN left over.

(a) He sells the pesos back to the bank. How much does he get in CAD?

(b) Those pesos cost him 2200×0.081 before the trip. How much money did he lose by changing them back?

11. Use the board from Question 9. Ella changes \$1200.00 CAD into US dollars. Her trip is cancelled, so that same day she changes all the US dollars back.

(a) How many US dollars does she get?

(b) How many Canadian dollars does she get back?

(c) How much did she lose?

(d) Explain why the bank's buying rate is lower than its selling rate.

12. Ravi needs \$400.00 USD in cash. Complete the table. Where should he buy it, and how much does he save compared with the most expensive choice?

Place	Rate and fee	Working	Total (CAD)
Bank	1 USD = 1.43 CAD, no fee		
Kiosk	1 USD = 1.40 CAD, plus \$14.95 fee		
Online order	1 USD = 1.38 CAD, plus 3% commission		

13. Noor wants new skates. A US website sells them for \$139.00 USD plus \$22.00 USD shipping. Her credit card uses 1 USD = 1.38 CAD and adds a 2.5% foreign transaction fee. A Calgary sports store sells the same skates for \$199.99 plus 5% GST. (Ignore any charges at the border.)

(a) Find the total cost of buying online, in CAD.

(b) Find the total cost at the Calgary store.

(c) Which is cheaper, and by how much?

(d) Give one reason, other than price, to buy skates at the local store.

14. You want to change \$60.00 USD into Canadian dollars. The rate is 1 USD = 1.38 CAD. Which calculation is correct?

A. $60 \div 1.38$ **B.** 60×1.38 **C.** $60 + 1.38$ **D.** 60×0.725

15. The rate is 1 USD = 1.38 CAD. The best estimate of the cost of a \$89.00 USD jacket in CAD is

A. \$65 **B.** \$90 **C.** \$120 **D.** \$150

16. A bank buys US dollars at 1.33 and sells them at 1.43 (CAD for 1 USD). Leon sells \$250.00 USD back to the bank after a trip. He gets

A. \$332.50 **B.** \$357.50 **C.** \$187.97 **D.** \$174.83

17. Numerical Response A hammock at a market in Mexico costs 1450 MXN. The rate is 1 MXN = 0.075 CAD. To the nearest dollar, the cost in CAD is _____.

18. Numerical Response A bank sells US dollars at 1.43 and buys them at 1.33 (CAD for 1 USD). Ben changes \$500.00 CAD into US dollars and then changes it all back on the same day. To the nearest dollar, he loses \$_____.

Lesson 11: Unit 1 Review

Assignment

This review is set up like the unit test. Show your work on the written questions. Round money to the nearest cent.

Part A: Written Response

1. Write each ratio in simplest form.
(a) 12 : 20 (b) 8 to 32 (c) 40 min : 1 h (d) 1.5 : 4.5

2. Find each unit price.
(a) A 4 L jug of milk for $6.29 (price per litre) (b) A dozen eggs for $4.68 (price per egg)

3. A restaurant bill in Red Deer is $46.80 before tax.
(a) Find the 5% GST. (b) Find the total with GST. (c) Find a 15% tip on the $46.80.

4. Solve each proportion.
(a) $\frac{x}{5} = \frac{12}{20}$ (b) $\frac{4}{9} = \frac{10}{x}$ (c) $7 : x = 21 : 36$

5. Amara earns $189.75 for 11 hours of work at a garden centre.
(a) What is her hourly rate of pay?

(b) How much would she earn in a 32-hour week?

(c) Her family's SUV used 47.6 L of gas to drive 560 km. What is its fuel economy in L/100 km?

6. Laundry detergent comes in three sizes. The number of loads is printed on each jug.

Jug	24 loads	52 loads	80 loads
Price	\$7.99	\$14.97	\$24.99
Price per load			

(a) Complete the table. Which jug is the best buy? _____

(b) Is the biggest jug always the best buy? Use the table to explain.

(c) Give one reason that is **not** about price for buying the 24-load jug.

7. Cans of soup are \$1.79 each, or 4 for \$6.00.

(a) What is the price of one can with the deal?

(b) How much do you save on 8 cans by using the deal?

8. An electronics store pays \$64.00 for a pair of headphones and sells them for \$99.99.

(a) What is the markup in dollars?

(b) What is the markup percent, to the nearest tenth of a percent?

(c) What does a customer in Alberta pay, with 5% GST?

9. Complete this receipt from a hardware store in Airdrie.

Item	**Quantity**	**Price each**	**Amount**
Box of deck screws	3	\$6.49	
Tube of caulking	2	\$8.79	
Drill bit set	1	\$24.99	
		Subtotal	
		GST (5%)	
		Total	

10. A TV regularly sells for $649.99. It is on sale for 20% off.

(a) What is the sale price?

(b) What is the total with GST in Alberta?

(c) What is the total in British Columbia (5% GST and 7% PST, 12% in all)?

(d) How much more does the TV cost in BC than in Alberta?

11. A blender usually sells for $40.00. One week the store raises the price to $50.00. The next week it advertises the blender at "20% off!"

(a) What is the percent increase from $40.00 to $50.00?

(b) What is the "sale" price?

(c) Is the sale really a deal? Explain.

12. T-shirts cost $24.00 each. A store offers two promotions, and you may use only one.
Promotion A: Buy 2, get the 3rd free. **Promotion B**: 30% off everything.

(a) Which promotion is cheaper if you want 3 T-shirts? Show both costs.

(b) Which promotion is cheaper if you want only 2 T-shirts?

(c) What does this tell you about choosing a promotion?

13. A bank shows these rates, in Canadian dollars for 1 unit of each currency.

Currency	**We buy**	**We sell**
US dollar (USD)	1.33	1.43
Euro (EUR)	1.47	1.57
British pound (GBP)	1.71	1.83

(a) Malik buys 250 euros in cash for a trip to Italy. How much does it cost him in Canadian dollars?

(b) He comes home with 35 euros and sells them back to the bank. How much does he get?

(c) Explain why Malik gets back less for each euro than he paid.

(d) His sister changes $500.00 CAD into British pounds at this bank. How many pounds does she get?

14. A pair of boots costs $119.00 USD on a US website, plus $18.00 USD shipping (assume no duty or other fees). The same boots cost $169.99 plus 5% GST in a Calgary store. Use 1 USD = 1.38 CAD.

(a) Estimate the US total in Canadian dollars in your head.

(b) Find the exact cost of each choice in Canadian dollars.

(c) Which is cheaper, and by how much? Give one other reason to buy in Calgary.

Part B: Multiple Choice

15. Which ratio is equivalent to 12 : 18?

A. 3 : 2 **B.** 2 : 3 **C.** 10 : 16 **D.** 6 : 12

16. A cyclist rides 54 km in 2.5 h. Her average speed is

A. 27 km/h **B.** 135 km/h **C.** 21.6 km/h **D.** 56.5 km/h

17. A chainsaw uses a 50:1 mix of gas to oil. How much oil goes with 2.5 L of gas?

A. 20 mL **B.** 50 mL **C.** 125 mL **D.** 0.05 mL

18. Cereal comes in 450 g for $4.99, 750 g for $7.49 and 1.2 kg for $12.99. The best buy is

A. the 450 g box

B. the 1.2 kg box

C. all three: they cost the same per 100 g

D. the 750 g box

19. A jacket's price drops from $80.00 to $60.00. The percent decrease is

A. 25% **B.** 20% **C.** 33.3% **D.** 75%

20. A store takes 30% off, then an extra 20% off the sale price. This is the same as a single discount of

A. 50% **B.** 56% **C.** 44% **D.** 6%

21. A bank's board shows US dollar: We buy 1.34, We sell 1.42. Which statement is true?

A. You pay $1.42 CAD for each US dollar you buy.

B. You pay $1.34 CAD for each US dollar you buy.

C. The bank pays you $1.42 CAD for each US dollar you sell back.

D. Changing CAD to USD and back again makes you money.

22. The exchange rate is 1 CAD = 0.72 USD. A \$45.00 USD shirt costs about how much in Canadian dollars?
A. \$32.40 **B.** \$62.50 **C.** \$45.72 **D.** \$57.60

Part C: Numerical Response

23. Numerical Response A cookie recipe uses 3 cups of flour for 24 cookies. The number of cups of flour needed for 60 cookies is _____.

24. Numerical Response A 680 g jar of peanut butter costs \$5.29. Its unit price, in **cents** per 100 g, to the nearest tenth of a cent, is _____ cents.

25. Numerical Response A shop buys a phone case for \$36.00 and sells it for \$49.50. The markup percent, to the nearest tenth, is _____%.

26. Numerical Response The exchange rate is 1 CAD = 0.66 EUR. The number of euros Noor gets for \$450 CAD, to the nearest euro, is _____.
